



NATIONAL ROCKETRY CHAMPIONSHIP 2025-26

Requirement Specification



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Document Information

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1 Acronyms & Reference Documents

1.1 Acronyms

Acronym	Description
ECSS	European Cooperation for Space Standardization
CAA	Civil Aviation Authority
UKRA	United Kingdom Rocketry Association
COTS	Commercial Off-the-shelf
SRAD	Student Research and Developed

1.2 Reference Documents

1.2.1 Competition Documents

There are two other documents related to the competition to make a total of **three regulatory documents**. The three documents are as follows:

ID	Document	Description
NRC-RB	Rulebook	Outlines the overall competition rules, funding, team structures and so on - the competition logistics.
NRC-SOW	Statement of Work	Outlines the main deliverables teams shall be expected to deliver - the competition project requirements.
NRC-RS	Requirements Specification (This Document)	Outlines the technical requirements the rocket must be designed to meet (Technical Specification)

1.2.2 Useful Reference Documents

No	Document	Description
1	ECSS-M-ST-10C Rev. 1	Space project management - Project planning and implementation
2	ECSS-S-ST-00-01C	Glossary of Terms - A list of commonly used terms in ECSS documents





2 Introduction

2.1 Scope

This document outlines the requirements for the National Rocketry Championship (NRC). This document contains two parts: NRC-RS-TSN (Technical Specification) and NRC-RS-MRD (Mission Requirements Document). Both parts adhere to the **ECSS (European Cooperation for Space Standardization)**, in order to teach participating teams the level of precision, formality, and traceability employed in industry.

The Technical Specification (RS-TSN) outlines the technical design requirements that all teams' launch vehicles must comply with. These requirements are informed by Health & Safety, UKRA rules and an obligation to maintain an acceptable baseline of engineering quality. Failure to meet these requirements may result in teams failing the Flight Readiness Review (FRR) and therefore being denied permission to launch.

The Mission Requirements Document (RS-MRD) is intended to guide participating teams towards a 'successful mission' on launch day. These requirements mirror practices in industry, where a "successful launch" differs from a "successful mission". This includes every requirement for the launch vehicle's flight, "The Customer's" payload and the rideshare payload. As mentioned in the statement of work (SOW), "The Customer's" payload will be provided by UKSEDS, but the rideshare payload should be provided by the participating team.





2.2 Requirement Numbering Definition

The following describes the layout of the requirements within this document.

NRC RS-TSN-**<A>**-**<n>** | Rev: **<R>**

<Title>

<text>

NRC RS-MRD-**<A>**-**<n>** | Rev: **<R>**

<Title>

<text>

VV Method: **<v>**

Parent: **<p>**

Where:

<RS-TSN> : Technical Specification

<RS-MRD> : Mission Requirements Document

<A>: Acronym for the section the requirement falls under.

<n>: Requirement number of 3 digits, starting from 001.

Note: **<A>** and **<n>** combine with the competition and document name acronyms to create the requirement ID.

<R>: Revision indicator, with "A" denoting the first version, "B" the second and so forth.

<Title>: Requirement title.

<text>: Requirement text.

<v>: the verification method(s) needed to demonstrate compliance to each requirement. The letters shall be only the following:

I: Inspection

D: Demonstration





A: Analysis

T: Test

<p>: Denotes if a requirement is derived from another requirement. Parent requirements shall be listed by the Requirement ID. If no Parent Requirement exists, the value shall be N/A.





◀NRC RS-TSN▶



1 Motors [MOR]

1.1 Mounting of the Motor to the Launch Vehicle

NRC RS-TSN-MOR-001 | Rev: A

The launch vehicle motor/motor case shall be clamped axially at both ends.

The teams must provide a means of securely mounting the motor within the launch vehicle. There are two types of motors that comply with NRC's rules: single-use motors and reloads. Single-use motors do not require a case, but reloads do. Therefore, teams should clamp a single-use motor directly or clamp the case for a reload.

The mount must transmit the thrust loads from the motor to the structure of the launch vehicle, provide axial alignment of the motor within the launch vehicle and prevent the motor from slipping out during handling and all flight phases.

An example motor mount arrangement is shown in Figure RS-1. The motor can be clamped at the motor thrust ring to the base of the launch vehicle. Please remember this configuration is just an example and may not work for every team's purposes.

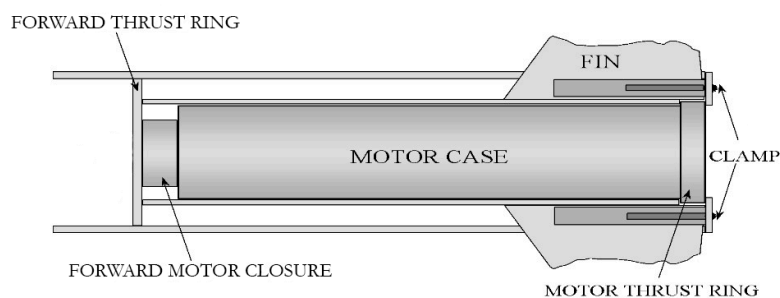


Figure RS-1: Motor mount arrangement schematic

VV Method: D

Parent: N/A





1.2 Motor Mount Strength Requirements

NRC RS-TSN-MOR-002 | Rev: A

The launch vehicle motor mount shall be designed to withstand an axial force equivalent to twice the maximum motor thrust without permanent deformation.

VV Method: T

Parent: N/A

NRC RS-TSN-MOR-003 | Rev: A

The launch vehicle motor mount shall be designed to withstand a lateral force in any direction equivalent to the maximum motor thrust with a misalignment of 5° cantilevered about the vehicle's centre of gravity without permanent deformation (yield).

The motor mount must be designed to take all the thrust loads, both axial and lateral, from the motor thrust ring (referred to in RS-1). The motor mount must be designed to withstand an axial force equivalent to twice the maximum motor thrust; as well as a lateral force in any direction, equivalent to the maximum motor thrust with an axial misalignment of 5° cantilevered about the vehicle's centre of gravity. The motor mount must then transmit these loads into the body tube without permanent deformation (yield).

VV Method: T

Parent: N/A





2 Stability Margin [SMN]

2.1 Motor Mount Strength Requirements

All launch vehicles participating in the NRC must comply with certain stability criteria. Normally, stability will be verified using slender body theory. For slender body theory to apply, the following four constraints are set on the vehicle:

1. Launch vehicles must have a length to diameter ratio (L/D) which lies between 12.5 and 35
2. The normal force coefficient C_N must be greater than 15 and less than 30.
3. The vehicle must fly at subsonic speeds.
4. The fins must have thin cross-sections.

If the basic criteria above cannot be met, further documentation must be presented to the RSO to demonstrate that the launch vehicle is both statically and dynamically stable.

2.2 Minimum Speed

NRC RS-TSN-SMN-001 | Rev: A

The launch vehicle shall have a minimum velocity of 20 m/s when leaving the launch rail.

At Midlands Rocketry Club NRC launch vehicles usually fly with 2m rails.

VV Method: A

Parent: N/A



2.3 Static Stability Margin

NRC RS-TSN-SMN-003 | Rev: A

The launch vehicle shall have a stability margin between 1.75 diameters and 2.5 diameters during all phases of flight before apogee.

The static stability margin is the distance between the centre of mass and the centre of pressure (see Figure RS-2). Please note that this is a static stability margin, i.e. when the launch vehicle is not in flight.

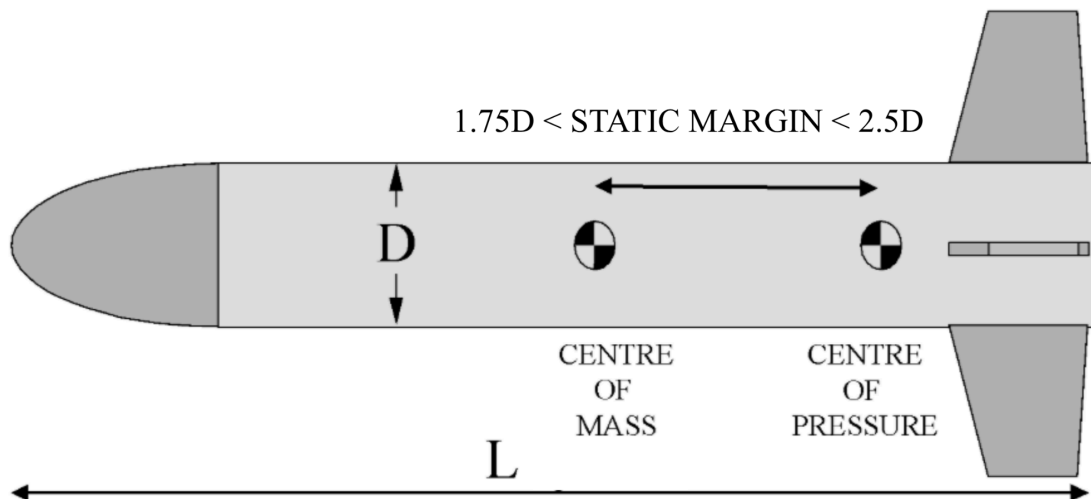


Figure RS-2: Schematic to illustrate the stability margin on a rocket

VV Method: A

Parent: N/A



3 Structural Acceptance Requirements [SAR]

For a launch vehicle to be accepted for flight it must pass certain pre-launch criteria, which include structural tests. Passing these tests will not guarantee that a launch vehicle will have a flawless flight but will ensure a minimum chance of failure and permit the launch vehicle to be safe for flight.

3.1 Fin Alignment

NRC RS-TSN-SAR-001 | Rev: A

The geometric alignment of each fin shall be within 2° of the projected longitudinal axes of symmetry of the launch vehicle.

See Figure RS-3 for more information.

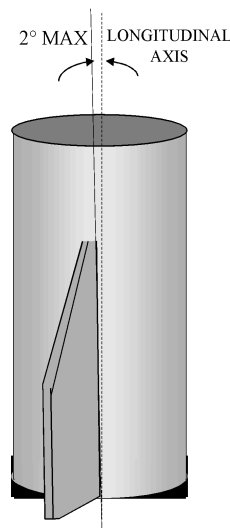


Figure RS-3: Schematic of to illustrate geometric alignment constraint of fins

VV Method: I

Parent: N/A



3.2 Fin Longitudinal Loading

NRC RS-TSN-SAR-002 | Rev: A

Each fin must be able to support a suspended load from its tip equivalent to twice the fin mass times the launch vehicle's maximum axial acceleration occurring during any flight phase without permanent deformation.

Please see **Figure RS-4** for more details.

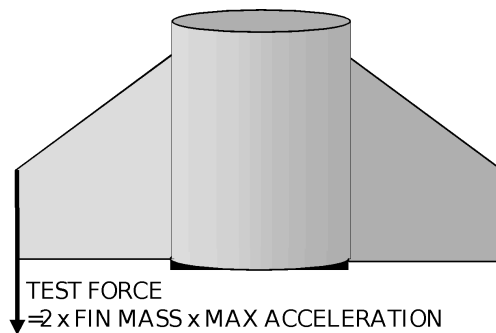


Figure RS-4: Schematic to illustrate of longitudinal loading force on individual fin

VV Method: T

Parent: N/A



3.3 Fin Lateral Loading

NRC RS-TSN-SAR-003 | Rev: A

Each fin shall withstand a transverse load in either direction, equivalent to the launch vehicle's launch mass when suspended from the fin tip with a maximum lateral deflection of less than 10° .

See Figure RS-5 for more information.

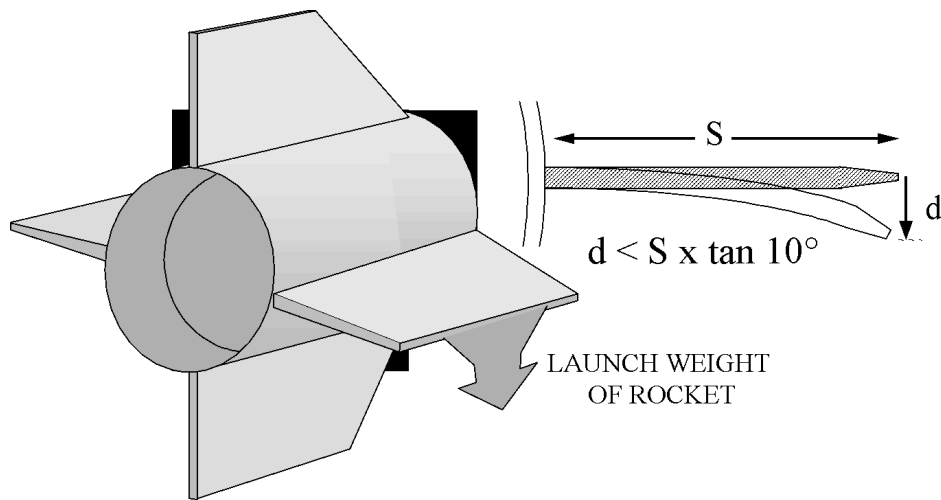


Figure RS-5: Schematic to illustrate of lateral loading force on individual fin

VV Method: T

Parent: N/A

3.4 Fuselage Stiffness

NRC RS-TSN-SAR-004 | Rev: A

The launch-ready launch vehicle shall deflect less than 10mm per metre length in any lateral direction when suspended from its centre of mass.

See Figure RS-6 for more information.

VV Method: T

Parent: N/A

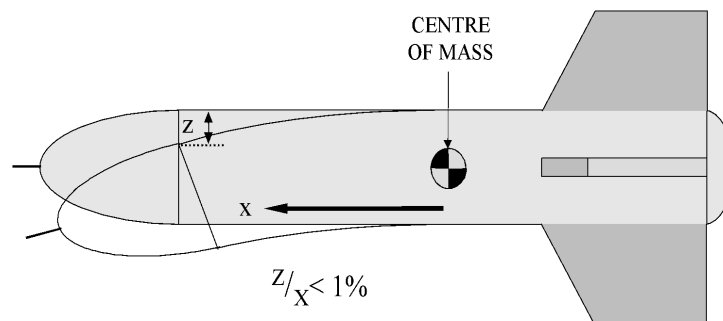


Figure RS-6: Schematic to illustrate fuselage stiffness of rocket

NRC RS-TSN-SAR-005 | Rev: A

The mating length between a coupler and body tube shall be a minimum of 1 body tube diameter.

When a launch vehicle is coupled together from several tubes, the method of joining the tubes is left to the application and discretion of the designer. However, it is required that the mating length between the coupler and each tube should be a minimum of 1 diameter when using plastics or composites (see Figure RS-7). This is advised to maintain satisfactory levels of stiffness along the length of the launch vehicle.

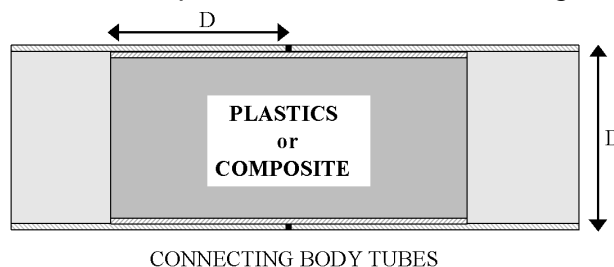


Figure RS-7: Schematic to illustrate mating length between coupler and body tube

For sliding connections, the minimum mating length between the coupler and tube should always be 1 diameter. Additionally, the fit between parts must stop any noticeable rotation at the joint.

VV Method: D

Parent: N/A

3.5 Static Strength Requirements

NRC RS-TSN-SAR-006 | Rev: A

All structural parts shall withstand twice their expected maximum inertial and aerodynamic loads without failure.

If participants wish to re-use their vehicles, all parts must withstand 150% of the applied loads without permanent deformation (yield).

VV Method: T

Parent: N/A



3.6 Launch Lug

NRC RS-TSN-SAR-007 | Rev: A

All launch vehicles shall be equipped with a minimum of 2 rail buttons.

At Midlands Rocketry Club we usually fly on 2m rails which are compatible with 6mm launch buttons. We recommend using Delrin rail buttons and an example can be found [here](#). A recommended layout of the launch buttons can be found in Figure RS-8.

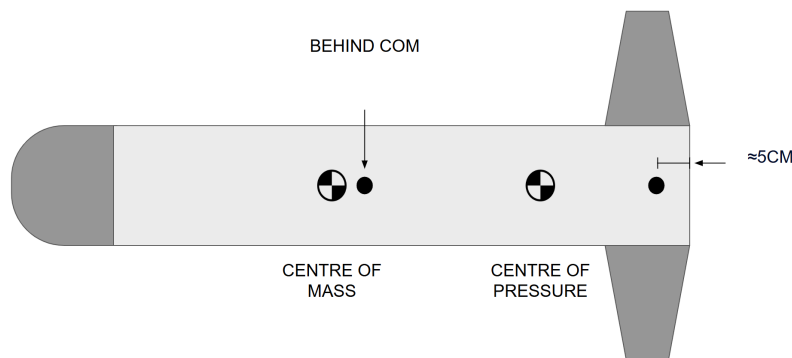


Figure RS-8: Schematic to illustrate recommended layout for launch lugs

VV Method: D

Parent: N/A



3.7 Finite Element Analysis

NRC RS-TSN-SAR-008 | Rev: A

Finite Element Analysis shall not be used to produce evidence for any structural requirements.

Whilst we would encourage teams to learn finite element analysis (FEA) for NRC and welcome its use in competition reports and documentation as supplementary evidence, for safety reasons, we cannot accept FEA evidence for verifying any structural requirements. We have seen teams in previous editions of NRC making incorrect assumptions, for example assuming homogenous material properties for additive manufacturing methods or 3D printing. These assumptions have led to issues at the flight readiness review where they find their design does not pass real-world structural tests. FEA is better suited to assessing trade-offs for material or design choices prior to physical testing occurring.

VV Method: D

Parent: N/A





4 Electrical Systems [ESS]

Teams may include several electrical subsystems onboard their launch vehicles, such as payloads, a telemetry system and an altimeter.

4.1 Internal Power Supply

NRC RS-TSN-ESS-001 | Rev: A

The launch vehicle shall be capable of being powered on and left to run using its internal power source for a minimum of 15 minutes on the launch pad.

Ideally, teams should have a way to arm the launch vehicle externally so that it can be armed on the launch pad. If this is not possible with the launch vehicle design chosen then the team will need to power it for much longer (up to an hour) on the competition day before it is placed on the pad. This should be budgeted for in addition to the energy used during the predicted total flight time.

VV Method: A

Parent: N/A

NRC RS-TSN-ESS-002 | Rev: A

The launch vehicle shall be capable of being armed at the launch site within 5 minutes.

Many delays on competition days are caused by teams having to fully disassemble their launch vehicles to arm them or switch on a payload. It is strongly advised that launch vehicles can be armed externally, this can be done with pull pins or screw switches for example.

VV Method: D

Parent: N/A





4.2 Electrical Considerations

NRC RS-TSN-ESS-003 | Rev: A

The launch vehicle shall not contain single-core wire.

Single-core wire shall not be used for electrical cable in primary systems since it is considered too fragile. Multi-core wire is tougher and more reliable under handling and flight conditions so this shall be used instead.

During assembly, testing and flight, electronic circuits and wiring are subjected to a high level of abuse. This requires that launch vehicle construction should be rugged, tidy and of a high standard of workmanship as possible.

Care should be taken when using electromechanical components (such as relays, switches, and connectors) to ensure that they are capable of withstanding high acceleration and vibration loads. Care should be taken to ensure that the switch is placed in the correct orientation. This is critical as they can only withstand high G loads along certain axes. Always refer to the manufacturer's data sheets.

The circuit must be structurally and electrically robust so that no parts of the circuit can change state or function due to any mechanical loads from transportation, manipulation on the launch pad or in flight.

Recovery system deployment shock loads are a very important design case and can exceed the thrust loads.

VV Method: D

Parent: N/A





4.3 Radio Frequencies

NRC RS-TSN-ESS-004 | Rev: A

Teams shall conform with Ofcom's IR 2030 radio frequency regulations.

A copy of Ofcom's IR 2030 can be found [here](#). All teams must ensure that they comply with these regulations when using radios.

Generally, teams are caught out with 915MHz which they will find is not permitted for airborne use. Depending on the specifics of the system, 868MHz and 433MHz are the most popular choices but please make sure to check the latest IR 2030 document.

VV Method: D

Parent: N/A





5 Recovery Requirements [RRS]

5.1 Use of Recovery System

NRC RS-TSN-RRS-001 | Rev: A

The launch vehicle shall have a recovery system.

All launch vehicles must have a system to recover them in a safe and controlled manner. This usually takes the form of a parachute that is activated when the launch vehicle reaches apogee.

VV Method: D

Parent: N/A

5.2 Landing Speeds

NRC RS-TSN-RRS-002 | Rev: A

The recovery system shall reduce the launch vehicle's vertical landing speed to less than 15 m/s.

This can be shown through OpenRocket simulations if teams use a parachute based recovery system. If teams use an alternative recovery system then they shall demonstrate this through hand calculations and possibly real-world tests.

VV Method: A

Parent: N/A

5.3 Maximum Post-Apogee Range

NRC RS-TSN-RRS-003 | Rev: A

After apogee, the launch vehicle shall not drift more than one kilometre before landing at wind speeds of 15m/s.

The prospective launch site is quite windy, but all launch vehicles must still land within the landowner's boundaries. It is advisable to make the recovery device highly visible to assist tracking.

VV Method: A

Parent: N/A





5.4 Motor Ejection Charge

Certain motors have an adjustable ejection charge which can be used to trigger the recovery system. This is a relatively passive approach, as the recovery system will deploy at a fixed time after burnout no matter what stage of flight the launch vehicle is in. As a result, premature and late deployment are common using this method.

Some delays are only adjustable to discrete time intervals from a maximum so teams have a limited amount of options for delay times e.g. 13 seconds, 10 seconds, 8 seconds, 6 seconds and 4 seconds.

Be wary of the delays on the selected motor and check that it is long enough to suit the time to apogee based on OpenRocket simulations.

5.5 Flight Computer

NRC RS-TSN-RRS-004 | Rev: A

Any flight computer used for recovery shall be a COTS or UKRA qualified system.

The purpose of a flight computer is to activate the recovery system when the launch vehicle reaches apogee.

An alternative method to using the motor's ejection charge is to use an electronic timer. This timer is activated on launch and is pre-programmed to wait for a set time interval before firing the recovery system. Some devices can be triggered by achieving a given acceleration, velocity or altitude. COTS means commercial-off-the-shelf. Essentially, this requirement says that teams cannot make their own flight computer and use it to trigger the recovery system, they must use a commercially available one. They may, however, make a flight computer and fly it as a payload, as long as it is not connected to the recovery system.

There is an exception to this requirement which is a SRAD (student researched and developed) or homemade flight computer that has been flight-qualified to UKRA's standards. If teams wish to develop their own recovery circuit, as per the UKRA safety code section 2, this will be classed as an experimental launch vehicle and so UKRA section 8 applies. This means that the **"homebrew" recovery electronics must be flown as a backup to a COTS recovery system** until they have been qualified as per section 8 of the UKRA Safety Code.

The UKRA Safety Code can be found [here](#).

VV Method: D

Parent: N/A





5.6 Isolation of Recovery Circuit

NRC RS-TSN-RRS-005 | Rev: A

Electronic recovery circuits shall be electrically isolated from other electronic systems.

It is recommended that all of the recovery sequencing circuitry (including the battery) be electrically isolated from any other electrical circuit used in the launch vehicle.

VV Method: D

Parent: N/A

5.7 The Launch Detector

Detection of the actual launch can be achieved using a variety of methods. These methods include the pulling out of a connector fixed to the launch pad, use of an optical sensor, use of a threshold accelerometer to initiate the time, or any other system a team wishes to develop (although these are not required).

5.8 Parachute Deployment

The best point to deploy the recovery device is at the instant the launch vehicle reaches apogee. This is when the speed of the launch vehicle is at its lowest and the recovery device deployment shock loads on the launch vehicles are the smallest. Normally, calculations are made using computer simulations to find the time from lift-off to apogee.

Recovery systems are usually triggered with the ejection charge built into the motor but sometimes other approaches are used such as separate black powder charges with a COTS flight computer.

5.9 Detection of Apogee

NRC RS-TSN-RRS-006 | Rev: A

Tilt switches shall not be used.

Detectors relying on the physical orientation of the launch vehicle relative to the gravity vector to detect apogee (such as tilt switches) are inaccurate and should not be used.





VV Method: D

Parent: N/A

5.10 Flight Computer Disarming Mechanism

NRC RS-TSN-RRS-007 | Rev: A

Any flight computers used to trigger recovery systems shall have an externally accessible disarming mechanism.

The flight computer (if used) must have a safe and secure disarming mechanism, which prevents inadvertent activation of the recovery system during handling and loading. This is especially important where pyrotechnic actuators are used.

The recovery system must be kept in the disarmed (safe) condition until the launch vehicle is safely loaded into the launch pad. At the designated point during countdown, the launch vehicle can then be armed.

VV Method: D

Parent: N/A

5.11 Transmission of Recovery Shock Loads

NRC RS-TSN-RRS-008 | Rev: A

Recovery loads shall be transmitted via eye bolts through bulkheads with a nut on the other side.

The main recovery shock loads must not be transmitted in shear through screw threads into the launch vehicle body as shown in Figure RS-9. It is recommended that these loads be transmitted through links and hook-eye anchor points.

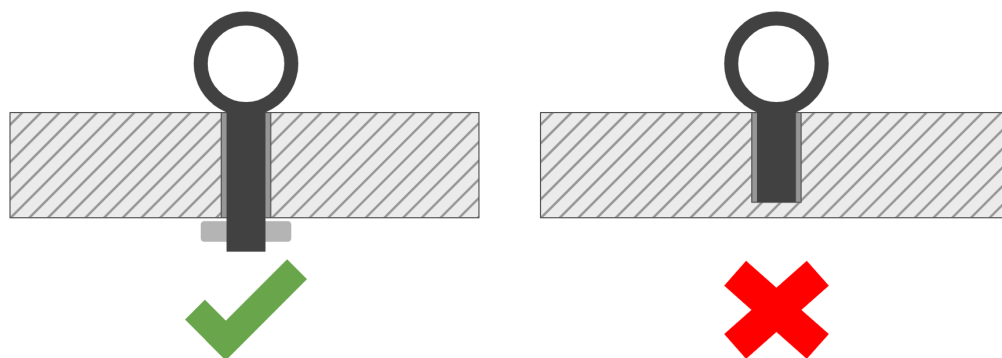


Figure RS-9: Schematic to illustrate the transmitted loads through bulkhead





VV Method: D

Parent: N/A

5.12 Shock Cord

NRC RS-TSN-RRS-007 | Rev: A

Recovery systems that use shock cords shall have a minimum length of 2m.

A lot of energy is released when the ejection charge fires. This energy must be great enough to separate the airframe and so if the shock cord is very short then there may not be sufficient cord length to dissipate this energy, leading to the shock cord snapping.

VV Method: D

Parent: N/A

NRC RS-TSN-RRS-008 | Rev: A

Any shock cords shall be made of Kevlar.

Previous NRC editions, as well as wider rocketry community flying, suggest that this is the best material to use for shock chords. It is also relatively inexpensive compared to alternative materials.

VV Method: D

Parent: NRC RS-TSN-RRS-007





6 Alternative Recovery [ARY]

6.1 Alternative Recovery Definition

NRC RS-TSN-ARY-001 | Rev: A

All launch vehicles shall conform to CAA UAS laws.

VV Method: D

Parent: N/A

NRC RS-TSN-ARY-002 | Rev: A

All launch vehicles shall conform with the UKRA Safety Code.

The UKRA Safety Code can be found [here](#).

VV Method: D

Parent: N/A

UKSEDS define alternative recovery methods as **any** method used to slow a launch vehicle down that does not use a conventional parachute.

The Civil Aviation Authority (CAA) and United Kingdom Rocketry Association (UKRA) have very strict laws and rules governing what can and cannot be done with rockets. To help teams adhere to them, the organisers have added the below sections as guidance.

Disclaimer: Please always refer to the CAA and UKRA's websites for up-to-date guidance and information. If you find any discrepancies between this document and their websites, please follow the CAA and UKRA's guidance and let the organisers know at: rocketry@ukseds.org





6.2 Alternative Deployment Methods

For any deployment that occurs **above** the CAA's drone limit of 120m/400ft either:

1. The ejection charge within the motor
2. A COTS (Commercial Off-The-Shelf) flight computer

must be used to trigger any deployment.

For the deployment of any Unmanned Aircraft Systems (UAS) from a launch vehicle, this **must** occur **below** the CAA's drone limit of 120m (400ft). The maximum distance a UAS should be flown from its operator is 500 metres or not beyond the visible line of sight if less than 500 metres.

More information about the CAA's drone laws can be found [here](#).

However, for safety reasons, the deployment of the launch vehicle's first stage recovery system must be triggered within 150m of apogee and the final stage (or the first stage if single stage) must deploy no lower than an altitude of 200m.

Any alternative recovery system designs proposed by a team in their reports will be submitted to the UKRA Safety and Technical Committee for approval before launch.

6.3 Active Stabilisation and Recovery

NRC RS-TSN-ARY-003 | Rev: A

Active stabilisation shall only be used on ascent.

As per the UKRA's safety code, active stabilisation can only be flown on ascent. This means that section 8 of their safety code **must** be followed.

Active recovery **must** only be used to guide the launch vehicle back to the launch site on the descent. It will also be classed as an experimental launch vehicle and so section 8 **must** also be followed.

The UKRA's safety code can be found [here](#).

VV Method: D

Parent: N/A





6.4 Altitude and Deployment Limits for UKRA Clubs

A list of active UKRA clubs can be found [here](#).

Please note that no person or team can break the max altitude of any site. This is set by the CAA and cannot be altered.

If your team is going to go above the dual deployment limit of any club, you must use a dual deployment system. For more information on dual deployment, please refer to Apogee Rocket's guide [here](#)



◀NRC RS-MRD▶

UKSEDS National Rocketry Championship 2025-26

Requirements Specification NRC-RS-001



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1. Mission Requirements [MIS]

NRC RS-MRD-MIS-001 | Rev: A

The launch vehicle should reach an apogee of 2200ft.

Below is a table outlining the mark distribution with respect to altitude.

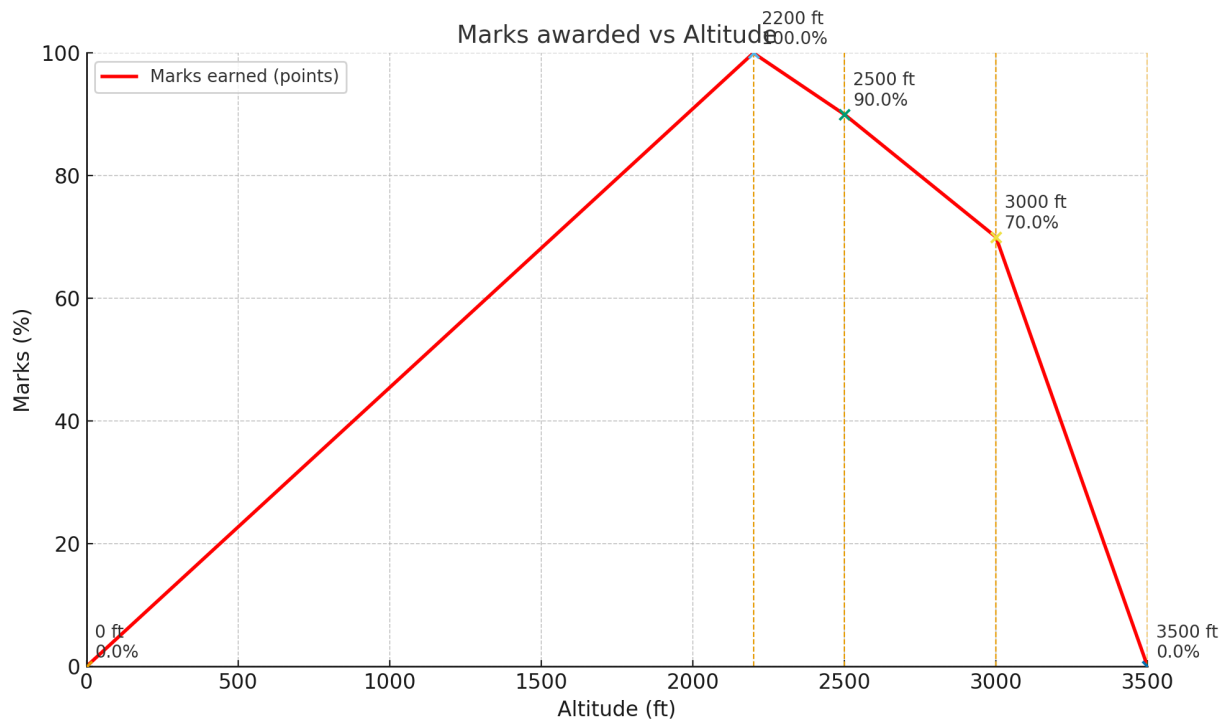


Figure RS-10 – Apogee vs Marks Awarded

VV Method: A

Parent: N/A





NRC RS-MRD-MIS-002 | Rev: A

The launch vehicle shall not fly above 3000ft without a dual-deployment recovery system.

Rockets achieving an apogee above 3000ft are required to incorporate a dual deployment recovery system onboard as per Midlands Rocketry Club rules. Rockets designed by teams for NRC are *not required* to incorporate dual deployment, due to the target apogee being 2200ft. As such, any rocket exceeding a 3000ft apogee will be severely penalised.

VV Method: A

Parent: NRC RS-MRD-MIS-001

NRC RS-MRD-MIS-003 | Rev: A

The launch vehicle shall not fly above the Midlands Rocketry Club's NOTAM limit.

NRC's launch site is overseen by the Midlands Rocketry Club, which can only secure a maximum altitude of 3500ft for the NRC competition day. Rockets are NOT permitted to fly above this altitude. This is a hard-and-fast rule. As such, any rockets breaching this maximum altitude will automatically receive 0 marks.

VV Method: A

Parent: NRC RS-MRD-MIS-001





NRC RS-MRD-MIS-004 | Rev: <A>

The launch vehicle shall be powered by a 2-grain 29mm G class ammonium perchlorate solid rocket motor with a total impulse under 130Ns.

Due to safety and competition restrictions, only approved motors can be used in the NRC. UKSEDS will not allow self-made motors or motors from unlicensed manufacturers. As standard, we approve [Cesaroni](#) and [AeroTech](#) motors as these are the standard ones used in the UK market. In previous NRC competitions we have had significant issues with communication between teams and the motor supplier. For this NRC edition, all teams shall order through UKSEDS so that the UKSEDS organising team is responsible for all communications with motor suppliers regarding motor orders for the competition day. This also ensures all of the motors are handled correctly by qualified persons with the correct licences.

Teams are limited to 2-grain 29mm Cesaroni reloads of G impulse and 29mm Aerotech G class motors with an impulse under 130Ns. These are classified as mid-power motors and so do not require UKRA certification to purchase or fly. For individuals wishing to obtain a UKRA level 1 certification, it is suggested that you look into UKSEDS' Project Odyssey. Please note that this is for individuals rather than teams due to the UKRA Level 1 certification process.

VV Method: D

Parent: N/A

NRC RS-MRD-MIS-005 | Rev: A

The rocket motor shall be supplied by Cesaroni Technology Incorporated or AeroTech.

Due to safety and competition restrictions, only approved motors can be used in the NRC. UKSEDS will not allow self-made motors or motors from unlicensed manufacturers. Motors from Cesaroni Technology Incorporated or AeroTech are UKSEDS verified and approved, and as such are the chosen suppliers for motors.

VV Method: D

Parent: NRC MRD-MIS-004





NRC RS-MRD-MIS-006 | Rev: A

The rocket motor shall be obtained through UKSEDS' NRC organising team.

In previous NRC competitions we have had significant issues with communication between teams and the motor supplier. For this NRC edition, all teams shall order through UKSEDS so that the UKSEDS organising team is responsible for all communications with motor suppliers regarding motor orders for the competition day. This also ensures all of the motors are handled correctly by qualified persons with the correct licences.

VV Method: D

Parent: NRC RS-MRD-MIS-004

NRC RS-MRD-MIS-007 | Rev: A

The launch vehicle shall carry a rideshare payload with a minimum mass of 100g.

The rideshare payload shall be provided by the teams. The nature of this rideshare payload shall be determined by the team.

HINT: some of the other requirements might indicate some features that your payload should incorporate but you may add more!

VV Method: D

Parent: N/A





NRC RS-MRD-MIS-008 | Rev: A

During descent, the launch vehicle shall drift a maximum horizontal distance of 750 m from the launch site under a windspeed of 10 m/s during descent.

NRC's launch site is surrounded by a number of private properties. In order to respect this and ensure easy recovery, the launch vehicle must land no farther than 750 m from the launch pad.

VV Method: A

Parent: N/A





2. "The Customer's" Payload [CPD]

Note: "The Customer's" payload will be provided on the day by UKSEDS. You are **not** expected to design or provide this. "The Customer's" payload has a power rating of 7.5W and a mass of 75g (not including the mass of the launch vehicle adapter).

NRC RS-MRD-CPD-001 | Rev: A

The launch vehicle shall provide a DC bus voltage of 3.5-5V to "The Customer's" payload.

VV Method: D,T

Parent: N/A

NRC RS-MRD-CPD-002 | Rev: A

The launch vehicle shall provide power to "The Customer's" payload for a minimum of 45 minutes.

This includes time on the launch pad, flight, and recovery. This requirement assumes that the launch vehicle is on the launch pad for 15 minutes.

VV Method: D

Parent: NRC RS-TSN-ESS-001

NRC RS-MRD-CPD-003 | Rev: A

The launch vehicle shall provide sufficient current to power "The Customer's" payload.

VV Method: D,T

Parent: N/A



NRC RS-MRD-CPD-004 | Rev: A

The launch vehicle shall provide power via a female XT30 connector at the top and bottom of the “The Customer’s” payload.

Please note: Either the top or bottom port can be used to provide power to the payload. Two options have been provided to give participants more flexibility in their designs.

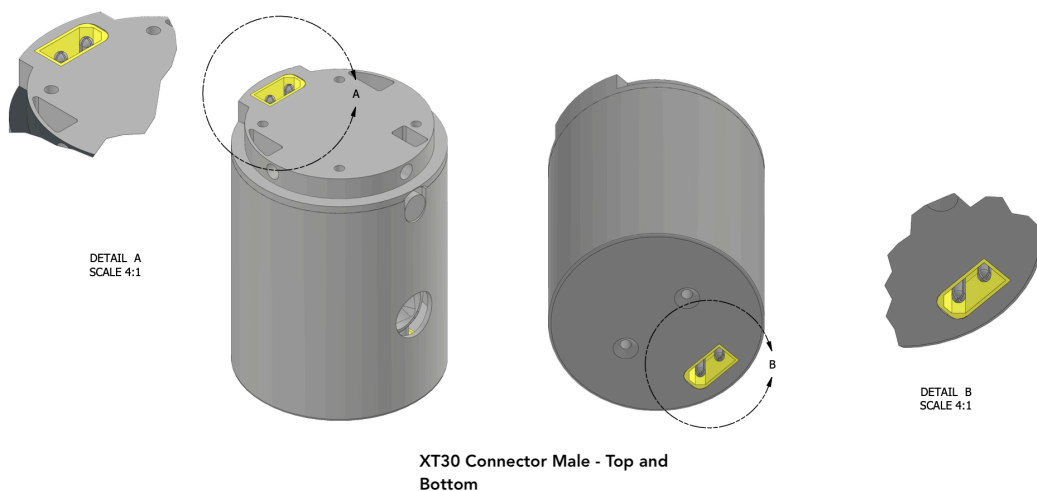


Figure RS-11: “The Customer’s” payload showing XT30 connector locations top and bottom

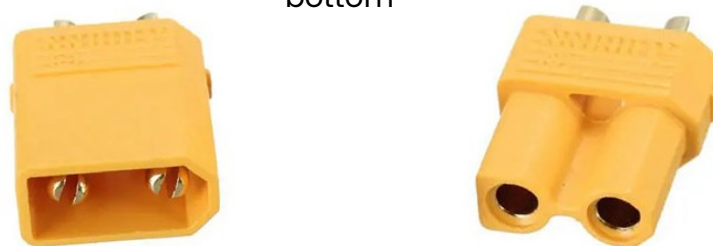


Figure RS-12 Male XT30 connector (left) to be provided by UKSEDS. Female XT30 connector (right) to be provided by participants.

VV Method: D

Parent: N/A





NRC RS-MRD-CPD-005 | Rev: A

The launch service provider shall provide a microSD with a minimum of 16GB storage for “The Customer’s” payload.

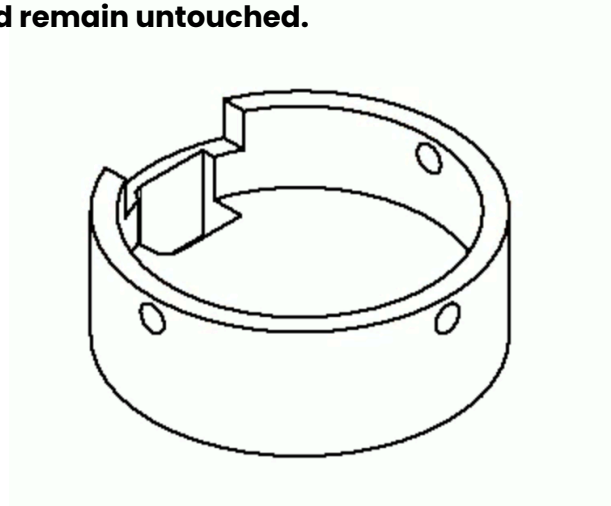
VV Method: D

Parent: N/A

NRC RS-MRD-CPD-006 | Rev: A

The launch vehicle shall mechanically interface with “The Customer’s” payload via the launch vehicle adapter.

The launch vehicle adapter CAD file has been shared with teams through the documentation page on the website. **Participating teams** are expected to manufacture and integrate the launch vehicle adapter in their designs. This has been done so that you design for and include the correct adapter in your design. It should also reduce interface issues on the competition day. **Teams are welcome to modify the launch vehicle adapter to interface with their designs. However the face that interfaces with the payload as well as the internal volumes and dimension of the ‘inner walls’ should remain untouched.**



Launch vehicle adapter

Figure RS-13 : Launch Vehicle adapter



VV Method: D, T

Parent: N/A

NRC RS-MRD-CPD-007 | Rev: A

The launch vehicle shall provide a minimum payload bay of 40mm diameter and 53mm length for “The Customer’s” payload (not including the launch vehicle adapter). Refer to Appendix A

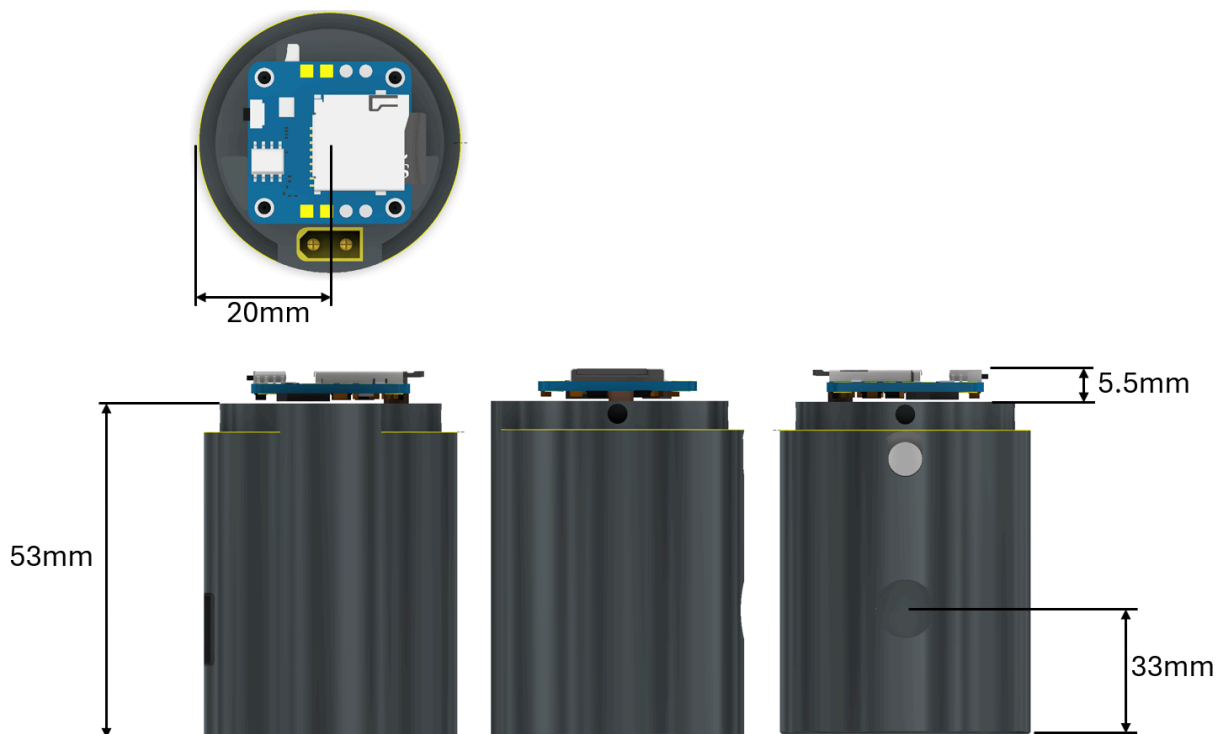


Figure RS-14 – Customer’s Payload

Important Note : Teams are advised to design their payload bays to have a small amount of extra space to account for any tolerancing issues at the integration stage. It is highly recommended to print out a copy of the dummy payload and carry out fit checks and work out an assembly procedure for the competition day.”

VV Method: D

Parent: N/A



NRC RS-MRD-CPD-008 | Rev: A

The thrust vector shall go through the geometric centre of "The Customer's" payload.

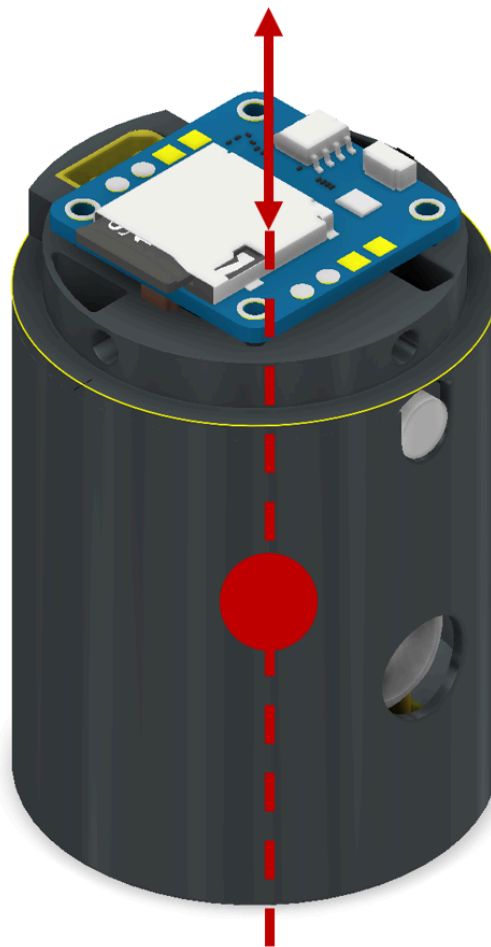


Figure RS-15 : Central Axis

VV Method: D

Parent: N/A

NRC RS-MRD-CPD-009 | Rev: A

The launch vehicle shall provide permanent external access to the following points for "The Customer's" payload (circled red). Refer to Appendix A

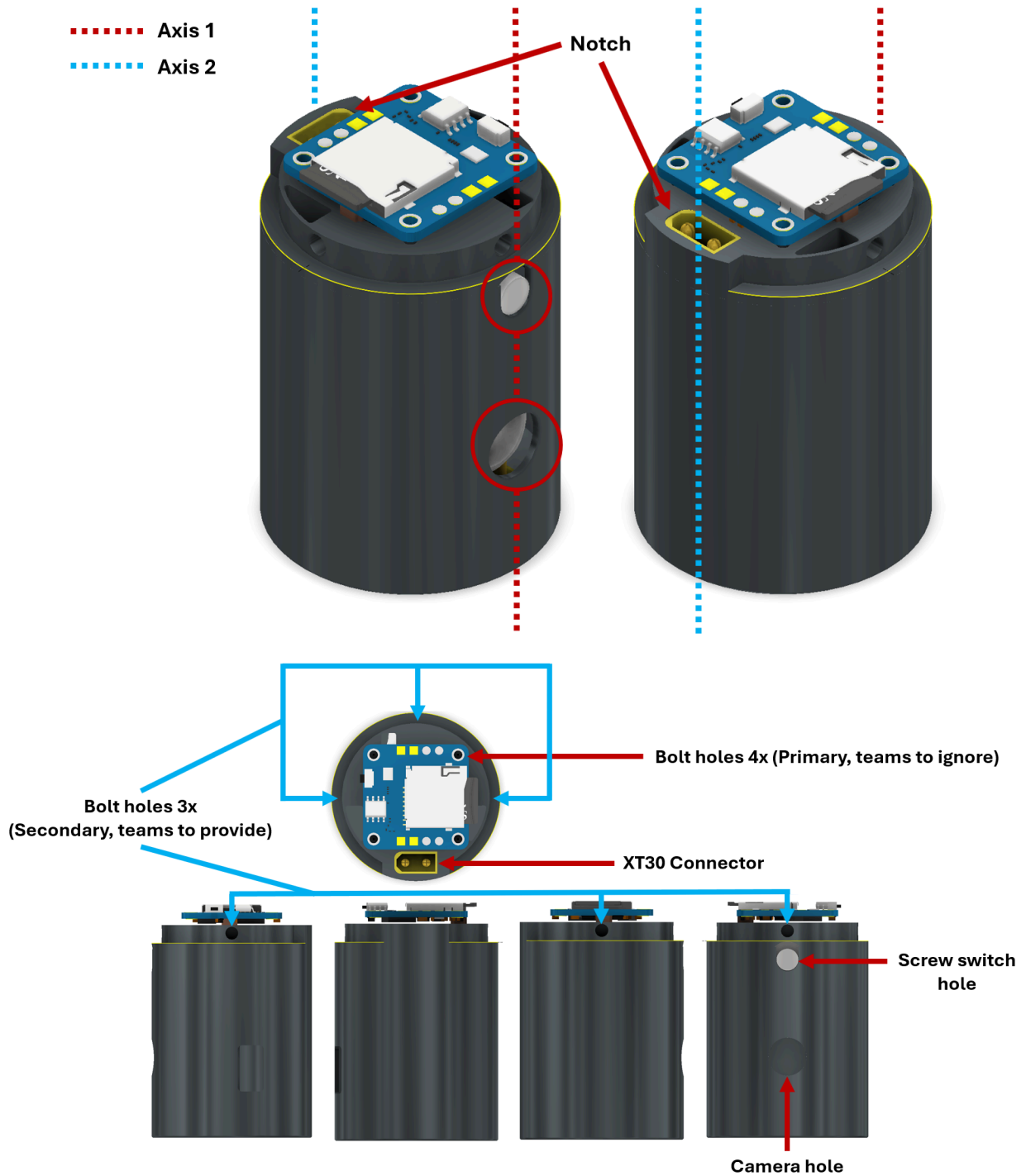


Figure RS-16: Customer Payload Alignment



VV Method: D

Parent: N/A

NRC RS-MRD-CPD-010 | Rev: A

The launch vehicle payload bay shall be unpressurised.

This allows "The Customer's" payload to read the altitude of its launch vehicle.

VV Method: D

Parent: N/A

NRC MRD-CPD-011 | Rev: A

Teams shall provide three M3 bolts to mount "the Customer's" payload to their launch vehicle adapter.

This is to "mate" the payload adapter and the payload itself. As standard, three M3 bolts are required. Teams are advised to bring spare bolts in a variety of lengths, in case they are dropped in the field at the launch site and are lost in the grass!

UKSEDS will have a small supply of spare M3 bolts in various sizes. We have added this requirement because many previous teams modified their launch vehicle adapter to use bolts of different lengths..

VV Method: D

Parent: NRC RS-MRD-CPD-006, NRC RS-MRD-CPD-009





3. Rideshare Payload [RPD]

NRC RS-MRD-RPD-001 | Rev: A

The rideshare payload shall contain a printed circuit board designed from scratch.

Teams should design, test and integrate a Printed Circuit Board (PCB) from scratch. If teams do not have possess prior experience with PCB Design, UKSEDS have provided a training course on using KiCAD to build your own PCB. This PCB should be designed dedicated for this year's competition. (Teams may not use a previous year's design).

VV Method: D

Parent: N/A

NRC RS-MRD-RPD-002 | Rev: A

The rideshare payload shall detect the launch vehicle's altitude.

IMU's are inclined to quickly drift over time and return inaccurate results. Therefore barometers are recommended in respect to this requirement.

VV Method: D

Parent: NRC RS-MRD-RPD-001

NRC RS-MRD-RPD-003 | Rev: A

The rideshare payload shall be able to produce an altitude vs time plot (in feet against seconds) within 10 minutes of recovery.

Note: the payload *does not require* a screen to display this plot. Teams may use an external screen such as a laptop if desired.

VV Method: D

Parent: NRC RS-MRD-RPD-001





NRC RS-MRD-RPD-004 | Rev: A

The rideshare payload shall be able to provide an apogee value in feet within 10 minutes of recovery without the use of external equipment.

Note: mechanical tools (screwdrivers, impact-drivers etc) are allowed to gain access to the rideshare payload. External digital equipment such as laptops to retrieve the data from the rideshare payload are prohibited.

VV Method: D

Parent: NRC MRD-RPD-001

NRC RS-MRD-RPD-005 | Rev: A

The rideshare payload shall include a silkscreen on the printed circuit board displaying the team logo.

VV Method: D

Parent: NRC RS-MRD-RPD-001

NRC RS-MRD-RPD-006 | Rev: A

The rideshare payload shall record at least one additional type of data and present it within 10 minutes of recovery.

Note: teams may use external equipment to display this data. This data must not be collected with the sensor used for NRC RS-MRD-RPD-002.

VV Method: D

Parent: NRC RS-MRD-RPD-001





END OF DOCUMENT

